

HaloHouse – Booking & Slot Management API (Technical Proposal)

This document outlines the proposed backend APIs required to support studio booking with date- and time-based slot selection. This is an API contract proposal only; no implementation details are included.

1. List Studios

GET /api/studios

Returns all available recording studios/spaces with basic pricing and type information.

2. List Services (Add-ons)

GET /api/services

Returns optional services that can be attached to a booking (pre-production, post-production, equipment).

3. Check Slot Availability

GET /api/availability

Query Parameters: studio_id, date (YYYY-MM-DD), duration (minutes)

Returns available start/end time slots for a given studio on a specific date.

4. Create Booking

POST /api/bookings

Creates a booking by locking an available slot. Requires studio ID, selected services, date, start time, end time, customer details, and timezone.

5. Retrieve Booking

GET /api/bookings/{booking_id}

Returns full booking details including studio, time window, services, and status.

6. Modify / Cancel Booking

PUT /api/bookings/{booking_id} – Reschedule booking

DELETE /api/bookings/{booking_id} – Cancel booking

Design Notes

- All bookings are slot-based (explicit date + start/end time).
- Availability API is the single source of truth to prevent double booking.
- Supports multiple services per booking.
- Timezone is included to support future multi-location expansion.
- API structure is designed to be frontend-agnostic and extensible.